Informed Consent

Exploring Dependency Management in the p5.js Ecosystem

Welcome and Thank You! We are a team of researchers from at VU Amsterdam and Université de Montreal, and we are currently conducting a study on dependencies within the p5.js ecosystem.

This survey is intended for p5.js developers, please only fill it out if you have experience with it. We are interested in understanding your perception of our proposed classification of dependencies present in **p5.js sketches**. For this study, you will be presented with information relevant to *dependencies* in p5.js sketches. Then, you will be asked to answer some questions about it.

The study contains 8 questions and should take you around 15-20 minutes to complete. Your participation in this research is voluntary. You have the right to withdraw at any point during the study.

This research is conducted in collaboration between the VU Amsterdam and the University of Montreal. If you have any comments or questions, please feel free to contact them via email.

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- Benoit Baudry (University of Montreal)
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Your responses will be kept completely confidential. Data collection and storage is GDPR compliant and will be saved anonymously.

By clicking the button below, you acknowledge: Your participation in the study is voluntary. You are 18 years of age or older. You are aware that you may choose to end your participation at any time and for any reason.

- ○ I consent, begin the study
- ○ I do not consent, I do not wish to participate

Background information

What is your main Occupation?

- ○ Artist
- ○ Web Designer
- ○ Creative Coder
- ○ Software developer
- ○ Student
- ○ Researcher
- ○ Other

Which of the following sectors best describes your field of work?

- ○ Creative industries
- ○ Research and education
- ○ Transportation
- ○ Communication
- ○ Government and public administration
- ○ Marketing and advertising
- ○ Media and entertainment
- ○ Science and technology
- ○ Finance and insurance
- ○ Retail

- ○ I do not use p5.js professionally (Hobby)
- ○ Other

How long have you been working with p5.js?

- ○ Less than a year
- ○ 1 to 3 years
- ○ 4 to 6 years
- ○ 7 to 9 years
- ○ 10 to 13 years
- ○ More than 13 years

Taxonomy of Dependencies

Generative art sketches often rely on external factors, ranging from libraries and custom assets to hardware capabilities, browser APIs, and runtime environments. We have developed a taxonomy to systematically categorize and analyze these dependencies to better understand how p5.js projects are built, maintained, and preserved.
Your feedback will help us validate and refine this taxonomy so it accurately reflects the real-world practices of the p5.js community.

Instructions

1. *Review the Taxonomy*: Below, you will find the proposed taxonomy, including each proposed category, and its definition.
2. *Evaluate Clarity & Completeness*: You will be asked to rate how clear the definitions are and whether any dependency types are missing.
3. *Insights*: Short text boxes will be available to suggest missing edge cases, highlight overlapping categories, or offer improvements.

P5 Sketches ~ Dependency Taxonomy

A browsable taxonomy of the dependencies that show up in p5.js sketches. Top-level categories start expanded; their sub-categories start collapsed. Click one to drill in or use the search box below to filter down to a dependency type.

We acknowledge that p5 offers different possibilities for generative art. However, the taxonomy below is focused on p5.js sketches that are displayed on a screen or projector. Therefore, physical computing aspects beyond the ones described below are not considered.

Search the taxonomy... Expand all Collapse all

Digital Assets

Media files the sketch loads and renders, such as images, audio, or 3D models.

- 3D meshes and models
- Audio samples
- Codebases
- Font files
- High-resolution textures
- Images and sprites
- Shader files
- SVG files
- Video files

Web API endpoints

External web services the sketch queries for live or third-party data, such as social feeds or weather.

- GraphQL APIs
- JSON APIs
- RESTful APIs
- SOAP APIs
- Web Sockets
- WebRTC
- XML APIs

Network protocols

Third-Party Libraries

How easy the categories are to understand	Not at all clear	Slightly clear	Moderately clear	Very clear	Extremely clear
How well the categories cover all p5.js dependency types	Not at all complete	Slightly complete	Moderately complete	Very complete	Extremely complete

What specific p5.js libraries, external assets, runtime/hardware dependencies, or code-level patterns are missing or unaddressed by this taxonomy?

Which specific categories did you find ambiguous, difficult to distinguish, or overlapping? Please share any suggestions for renaming or regrouping them

How confident would you feel applying this taxonomy to categorize or audit dependencies in your own p5.js projects?

- ○ Not at all confident
- ○ Slightly confident
- ○ Moderately confident
- ○ Very confident
- ○ Extremely confident

Did you notice any overlap where a single dependency type could fit into multiple categories simultaneously?

- ○ Yes
- ○ No
- ○ Unsure

Which categories overlapped, and under what circumstances?

We want to ensure that the proposed taxonomy is as useful and comprehensive as possible. p5.js projects often involve unique setups (e.g., WebGL shaders, node packages, hardware inputs like webcams/MIDI, or iframe embedding).
Based on your experience developing with p5.js, *are there edge cases in your workflow that this taxonomy fails to accommodate*?

Final

Do you have any final thoughts, suggestions, or experiences regarding p5.js dependencies that we haven't covered? We'd love to hear your feedback.